

AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework

SECURITY ASSESSMENT REPORT

Report ID:	REPORT-20260930-085431-DEECC5	Report Type:	EXECUTIVE
Generated At:	2026-09-30 08:54:31 UTC	Scope:	Entire Telemetry
Application Mode:	STANDALONE	Architecture Status:	10 of 10 Layers Active
Backend Status:	OPERATIONAL	Database Status:	CONNECTED

ASSESSMENT OBJECTIVE & VERIFICATION NOTICE:

This formal security report consolidates empirical findings produced by the framework's protocol analyzer, SA state machine, behavioral drift detector, Isolation Forest anomaly engine, and deterministic security rule catalog. Every metric and finding is grounded directly in captured network telemetry and registered security rules without synthetic scores.

1. Executive Summary & Security Posture

Total Packets	Active SAs	Vulnerabilities	ML Anomalies	Drift Events
1419	8	9 (0 Crit, 4 High)	0	0

RISK ASSESSMENT ENGINE (LAYER 10) STATUS: OPERATIONAL (IDLE)
Layer 10 (Risk Assessment & Decision Engine) is operational. No session risk assessments have been committed yet for this scope.

1A. Security Assessment Engine — Cryptographic & Configuration Posture

SESSION IPSEC-B12F4A51BDC0 — SECURITY SCORE 48.2 / 100 (CRITICAL_DEFICIENCIES)
Evidence coverage: 6 of 6 components backed by evidence (100% of scoring weight). | **AI confidence:** 91%
Protocol: ESP+IKE | **IKE:** 1.0 | **Mode:** TUNNEL [ASSUMED] | **Traffic inside ESP:** ICMP (96%) [PREDICTED]

Dimension	Result	Provenance	Evidence
Cryptographic strength	Grade F — 3DES / MD5 / MODP-1024 (Group 2)	OBSERVED	Weakest-link security strength ≈ 64 bits (NIST SP 800-57 equivalence) for 3DES / MD5 / MODP-1024 (Group 2).; Deprecated transform(s) in the negotiated suite: 3DES, MD5, MODP-1024 (Group 2).
Forward secrecy	DISABLED (INSECURE, 55%)	INFERRED	Child SA rekeys reuse the IKE SA keying material without PFS (inferred, 55%); compromise of the IKE SA keys allows retroactive decryption of recorded ESP traffic. IKEv1 Quick Mode messages (packets 4, 5, 6) are only 212 bytes: no room for a KE payload, so Phase 2 keys derive from the Phase 1 SA without PFS.
Replay protection	PROTECTED — 0 duplicates, 0 reordered (max distance 0)	OBSERVED	No reordering observed; any window ≥ 32 packets (RFC 4303 minimum) suffices.
Key lifetime	EXCEEDED — 29s / 10238 bytes; negotiated 172800s	OBSERVED	Observed duration 29.4s and volume 10238 bytes are within the 8 h / 4 GB Child SA benchmark. 2 rekey exchange(s) observed. IKEv1 negotiated an IKE SA lifetime of 172800s, above the 24 h benchmark.
Metadata exposure	MEDIUM (26.6/100) — TFC padding detected	PREDICTED	spl_leakage_score: 45.0/100; sequence_leakage_score: 30.0/100; packet_length_leakage_score: 12.0/100; timing_leakage_score: 28.0/100; topology_leakage_score: 20.0/100
Rule compliance	NON_COMPLIANT — 16/24 rules passed	OBSERVED	RULE-AUTH-002 (MEDIUM); RULE-CONF-002 (MEDIUM); RULE-CRYPTO-001 (HIGH); RULE-CRYPTO-002 (HIGH); RULE-CRYPTO-003 (HIGH); RULE-CRYPTO-007 (LOW)

Compliance profile	Status	Score	Coverage	Summary
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IETF baseline (RFC 8247 / RFC 8221 / RFC 9395)	NON_COMPLIANT	0	67%	4 of 4 verifiable requirement(s) violated. Coverage 67%: 2 requirement(s) could not be verified from the capture.
NIST SP 800-77 Rev 1 (with SP 800-131A Rev 2 transitions)	NON_COMPLIANT	0	71%	5 of 5 verifiable requirement(s) violated. Coverage 71%: 2 requirement(s) could not be verified from the capture.
NSA CNSA Suite 1.0 (IPsec profile)	NON_COMPLIANT	0	71%	5 of 5 verifiable requirement(s) violated. Coverage 71%: 2 requirement(s) could not be verified from the capture.

Priority findings: Deprecated Cipher Detected (3DES) (HIGH); Weak Integrity Algorithm (MD5) (HIGH); Insufficient Diffie-Hellman Group (MODP-1024 (Group 2)) (HIGH); IKEv1 Aggressive Mode Cleartext Hash Risk (HIGH)

8. AI Encapsulated Traffic-Type Classification (ESP Inner Flow Inference)

Total Evaluated Sessions: 0 · Distribution: N/A

Session ID	Traffic Type	Confidence	AI Feature Attribution & Explainability
IPSEC-B12F4A51BDC0	ICMP	95.8%	None: small=1.00, ratio=0.97; None: iat=0.50s, cv=0.89; None: 2.0

12. Prioritized Remediation Recommendations

Priority	Recommendation Action	Implementation Details
High Priority	Remediate High Risk: Deprecated Cipher Detected (3DES)	1. Deprecate legacy ciphers (DES, 3DES, Blowfish, RC4, NULL) immediately. 2. Reconfigure the IPsec peer proposal to require AEAD algorithms: AES-256-GCM or ChaCha20-Poly1305 (RFC 8221 MUST recommendation). 3. Where AEAD is unavailable, configure AES-256-CBC with HMAC-SHA-256 integrity. 4. Enforce strict cryptographic suite compliance across both IKE SA and Child SA proposals.
High Priority	Remediate High Risk: Weak Integrity Algorithm (MD5)	1. Remove MD5, SHA-1, and truncated HMAC-MD5-96 from IKE and ESP integrity proposals. 2. Transition to SHA-2 family integrity algorithms: HMAC-SHA2-256-128, HMAC-SHA2-384-192, or HMAC-SHA2-512-256. 3. Prefer authenticated encryption (AES-GCM) where integrity is integrated cryptographically.
High Priority	Remediate High Risk: Insufficient Diffie-Hellman Group (MODP-1024 (Group 2))	1. Disallow legacy Diffie-Hellman groups 1 (768-bit), 2 (1024-bit), and 5 (1536-bit) which fail to provide adequate bit-strength against modern cryptanalysis. 2. Configure modern DH groups: Group 14 (2048-bit MODP), Group 19 (256-bit ECP / NIST P-256), Group 20 (384-bit ECP), or Group 21 (521-bit ECP). 3. For modern deployments, prefer Curve25519 (Group 31) per RFC 8221.
High Priority	Remediate High Risk: IKEv1 Aggressive Mode Cleartext Hash Risk	1. Discontinue IKEv1 Aggressive Mode immediately. 2. Aggressive Mode sends the initiator authentication hash in the clear across packets 1 and 2, exposing pre-shared keys to offline dictionary and rainbow table cracking. 3. Upgrade to IKEv2 or restrict legacy peers to Main Mode with complex certificates.

Medium Priority	Harden Configuration: Cleartext Peer Identity Leakage	1. Upgrade the VPN endpoint configuration to IKEv2 (RFC 7296). 2. IKEv1 is formally deprecated by RFC 9395 due to protocol-level design vulnerabilities (lack of DoS resilience, complex multi-mode negotiations, cleartext identities). 3. Disable IKEv1 fallback on the gateway daemon to prevent downgrade attacks.
Medium Priority	Harden Configuration: Child SA Rekey Without Perfect Forward Secrecy	1. Perfect Forward Secrecy (PFS) ensures that compromise of long-term peer keys does not decrypt past recorded session traffic. 2. Enable PFS on Child SA proposals by requiring a new Diffie-Hellman exchange during CREATE_CHILD_SA / Quick Mode. 3. Select DH Group 14 or higher for PFS key derivation.
Medium Priority	Harden Configuration: Traffic Flow Confidentiality (TFC) Inactive	1. Traffic Flow Confidentiality (TFC) per RFC 4303 Section 2.7 conceals true packet payload lengths. 2. Enable TFC padding on security gateways to pad frames to fixed quantum blocks (e.g. 128B or 256B). 3. Inject periodic dummy packets during idle intervals to obscure traffic presence and bursting.
Medium Priority	Enable Perfect Forward Secrecy (PFS)	Configure Diffie-Hellman group in Child SA phase 2 negotiations to guarantee forward secrecy.

Data Provenance & Telemetry Integrity Audit

In strict compliance with NTRO / SIH 26160 cybersecurity requirements, the framework transparently categorizes all report data points into: **Observed** (verifiable packet telemetry), **Inferred** (derived from handshake/transforms), **Predicted** (machine-learning classification), and **Unavailable** (protected by ESP encryption or mid-session capture start).

Telemetry Element	Data Classification	Epistemological Basis
Packet Headers	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Ip Endpoints	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Security Parameter Indices	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Sequence Numbers	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Traffic Cadence And Timing	OBSERVED	Directly extracted from raw PCAP byte streams and network interface frames.
Vpn Encapsulation Mode	INFERRED	Logically deduced from unencrypted IKE handshake proposals and header metadata.
Cryptographic Algorithms	INFERRED	Logically deduced from unencrypted IKE handshake proposals and header metadata.
Diffie Hellman Group	INFERRED	Logically deduced from unencrypted IKE handshake proposals and header metadata.
Encrypted Traffic Application Type	PREDICTED	Estimated via statistical machine learning and behavioral traffic flow classification.
Ai Anomaly Attack Likelihood	PREDICTED	Estimated via statistical machine learning and behavioral traffic flow classification.
Cleartext Inner Payload	UNAVAILABLE	Protected by ESP payload encryption or missing due to mid-session capture start.
Unobserved Rekey Proposals	UNAVAILABLE	Protected by ESP payload encryption or missing due to mid-session capture start.

13. Technical Appendix & Architecture Audit

This document is an automated cybersecurity assessment report generated by the AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework. Findings reflect actual network telemetry, deterministic rule evaluation, and empirical machine-learning observations.

#	Architecture Layer	Package	Status
1	IPsec VPN Test Environment	layer01_test_environment	PARTIALLY_OPERATIONAL
2	Packet Capture & Data Collection	layer02_packet_capture	PARTIALLY_OPERATIONAL
3	Packet & Protocol Analysis	layer03_protocol_analysis	OPERATIONAL
4	Security Association & Protocol State Analysis	layer04_sa_lifecycle	OPERATIONAL
5	Feature Extraction & Engineering	layer05_feature_engineering	OPERATIONAL
6	IPsec Session Fingerprinting	layer06_session_fingerprinting	OPERATIONAL
7	AI-Based Protocol & Traffic Classification	layer08_ai_ml	OPERATIONAL
8	Security Assessment Engine	layer09_vulnerability_engine	OPERATIONAL
9	Risk Assessment & Decision Engine	layer10_risk_engine	OPERATIONAL
10	Dashboard & Report Generation	layer14_reports	OPERATIONAL